files with this Ruby installation," both of which we recommend. You'll then have the option to install "Ruby RI and HTML documentation" and the "MSYS2 development toolchain," which again, we recommend. At the end, you'll be asked to run "ridk install" to set up the development toolchain. Doing so will give you a popup window that will ask you which MSYS2 components to install and to confirm the defaults are what you want (which they are, keep the defaults). Press ENTER to start the MSYS2 installation. After the installation finishes that phase, it will prompt you again. If the brackets in the prompt are empty, pressing ENTER will finish the installation.

### Using Ruby With RubyInstaller

At this point from a regular Windows terminal, you can run `ruby` and `irb` with the same options as we discussed in Chapter 14, Ruby from the Command Line, on page 231 and Chapter 16, Interactive Ruby, on page 277.

You'll find two versions of the Ruby interpreter in the RubyInstaller distribution. The `ruby` version is meant to be used at a command prompt (DOS shell or PowerShell), just as in the Unix version. For applications that read and write to the standard input and output, this is fine. This means that any time you run `ruby`, you'll get a DOS shell even if you don't want one—Windows will create a new command prompt window and display it while Ruby is running.

This may not be appropriate behavior if, for example, you double-click a Ruby script that uses a graphical interface or if you are running a Ruby script as a background task or from inside another program. In these cases, you will want to use `rubyw`. It is the same as `ruby` except that it does not provide standard in, standard out, or standard error and does not launch a DOS shell when run.

On Windows 11, you can also run Ruby code by right-clicking Ruby files in the file explorer.

## Running Ruby

Now that Ruby is installed, you'd probably like to run some programs. Unlike compiled languages, you have two ways to run Ruby—you can type in code interactively, or you can create program files and run them. Typing in code interactively is a great way to experiment with the language, but for code that's more complex or code that you will want to run more than once, you'll need to create program files and run them. But, before we go any further, let's test to see whether Ruby is installed. Bring up a fresh command prompt, and type this:

```
$ ruby --version
ruby 3.2.0 (2022-12-25 revision a528908271) [arm64-darwin22]
```

Technically, you can run Ruby interactively by just typing `ruby` at the shell prompt. You'll get a blank line in response, and you can type Ruby code there.

```
$ ruby
puts "Hello, world!"
^D
Hello, world!
```

In this example we typed in a single line of Ruby. That line consists of two parts. The first part, `puts`, is the name of a *method*. A method is a pre-defined chunk of code. In this case, the `puts` method is one of several methods defined for us by Ruby. The second part, `"Hello, world!"`, is text surrounded by double quotes, which is called a *string*. Combining the two, the Ruby